Notes - Review on Qubit Physical Realization Methods [1]

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Introduction

This paper covers four methods to physically realise qubits:

1. Superconducting Qubit
2. Trapped-Ion Qubit
3. Photonic Qubit
4. Spin-based Qubit

Math Tools of Physical Realisation of Qubits

This section mainly just introduces the Hamiltonian, the Schrödinger Equation, and bra-ket notation, all of which we already know.
However, it also makes a statement I wasn't aware of – A qubit is defined as a system restricted to the two lowest energy levels of an anharmonic potential, rather than a free particle moving in continuous space.
It is obvious that a qubit cannot be a free particle, as it needs to be discrete and cannot have a continuous energy spectrum like that of a free particle. But why anharmonic?
The problem with harmonic oscillators is that all energy levels have the same gap between them. If the energy eigenvalues are $\hbar\omega + \frac{1}{2}$, then a drive of frequency ω will cause the $|0\rangle \to |1\rangle$ transition, the $|1\rangle \to |2\rangle$ and all other consecutive transitions. So you cannot get any specificity.
The anharmonic oscillator, on the other hand, has unevenly spaced energy levels. So one is able to tune the drive frequency such that only the $|0\rangle \to |1\rangle$ transition is caused. These two states are the "two lowest energy levels" mentioned in the paper.

1 Superconducting Qubit

Here the qubits are made using electronic circuits composed of superconducting materials, such as aluminium, which enter a distinct physical state characterised by zero electrical resistance when cooled below its critical temperature T_C, as shown in Figure 1.

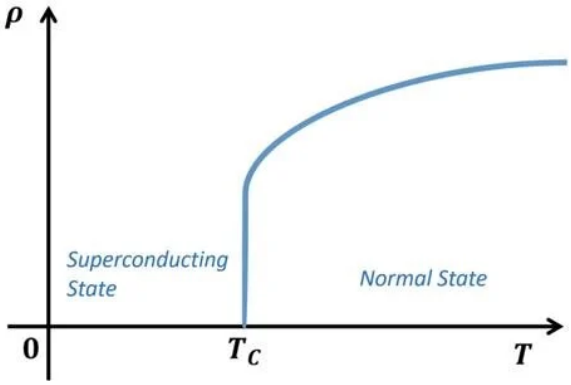


Figure 1: Resistivity vs. Temperature of a superconductor [1]

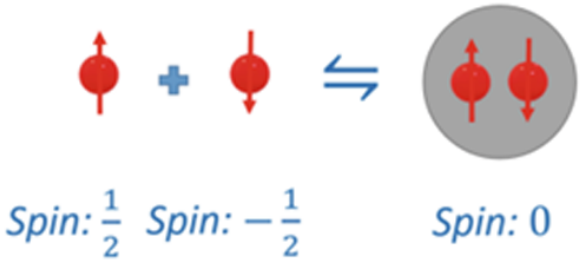


Figure 2: Pair of electrons making a 0-spin Cooper Pair [1]

1.1 How Superconductors Work

The phenomenon of superconduction arises because at low temperatures, electrons in the superconductors condense into bound pairs known as *Cooper Pairs*, as shown in Figure 2. These electrons are able to stay in pairs, despite Coulomb forces, thanks to the slightly positively-charged region surrounding an electron as it moves. At low temperatures, this attraction holds for long enough to have an effect, thus forming Cooper pairs.
These pairs have 0 spin, meaning they are bosons (integer spin). Since multiple bosons can occupy the same state (unlike fermions), all these pairs can condense into the same quantum state simultaneously. Hence, the microscopic behaviour of these electrons condenses to a *macroscopic quantum state*. Since all pairs share the same state, scattering one pair out of the state will require much more energy. Δ is defined as the *superconducting energy gap*, which is the energy difference between an electron in a cooper pair and an individual electron. For a standard Aluminium junction, this gap is directly related to the critical temperature by the Bardeen-Cooper-Schrieffer (BCS) Limit:

$$\Delta(T=0) = 1.764k\,T_C \tag{1}$$

Now to break a Cooper pair into two electrons, we would require 2Δ energy, which cannot be provided at such low temperatures ($T < T_c$). And since there is no scattering, flowing current experiences no dissipation, and the material exhibits *zero resistance*.
The collective state of the material is decribed by

$$\psi(t) = \sqrt{n}\,e^{i\varphi(t)} \tag{2}$$

where n represents the density of Cooper pairs and $\varphi(t)$ denotes the superconducting phase.

1.2 The Josephson Junction

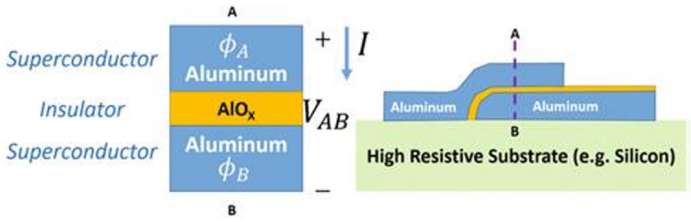


Figure 3: Josephson Junction [1]

The component that enables quantum processing using superconductors is the *Josephson junction*, depicted in Figure 3. It is made of two superconductors A and B, with a thin insulator sandwiched between them (forming the "junction"). The insulator is thin enough to allow for *quantum tunnelling* between the superconductors.
The wavefunctions of the two superconducting regions will be

$$\begin{aligned}\psi_A(t) &= \sqrt{n_A}\,e^{i\varphi_A(t)} \\ \psi_B(t) &= \sqrt{n_B}\,e^{i\varphi_B(t)}\end{aligned} \tag{3}$$

The phase difference between these regions will be given by $\varphi(t) = \varphi_A(t) - \varphi_B(t)$.
Now, thanks to this thin insulator layer, the TDSE's of the two superconducting regions will be coupled in some way. Upon solving the real part of the coupled DE's, one can get the rate of change of Cooper pair concentration, which indicates the tunnelling current $I(t)$. After solving, it will have the form

$$I(t) = I_C \sin(\varphi(t)) \tag{4}$$

where I_C is the critical current of the junction.
Now, taking the imaginary part of the two coupled DE's and subtracting them, we get the rate of change of phase $d\varphi(t)/dt$.

$$\frac{d}{dt}\varphi(t) = \frac{2e}{\hbar}V_{AB} = \frac{2\pi}{\phi_0}V_{AB} \tag{5}$$

where V_{AB} is the potential difference across the junction. ϕ_0 is the *magnetic flux quantum*, the fundamental unit of quantisation of magnetic flux in quantum systems, particularly in superconductors. It has the value $h/2e = 2\pi\hbar/2e$. Now we solve this DE

$$\begin{aligned}
\varphi(t) - \varphi(0) &= \int_0^t \frac{2\pi}{\phi_0}V_{AB}\,dt' \\
\Rightarrow \varphi(t) &= \varphi(0) + \frac{2\pi}{\phi_0}V_{AB}\,t \\
\Rightarrow I(t) &= I_C \sin\left(\varphi(0) + \frac{2\pi}{\phi_0}V_{AB}\,t\right)
\end{aligned} \tag{6}$$

If we differentiate (4) on both sides, we get

$$\begin{aligned}
\frac{dI}{dt} &= I_C\cos(\varphi(t))\frac{d\varphi}{dt} \\
&= I_C\cos(\varphi(t))\frac{2\pi}{\phi_0}V_{AB} \\
\Rightarrow V_{AB} &= \frac{\phi_0}{2\pi I_C}\frac{1}{\cos(\varphi(t))}\frac{dI}{dt}
\end{aligned} \tag{7}$$

Comparing this to the $V - I$ relation of an inductor, one can see that the Josephson junction behaves as a non-linear inductor, with inductance given by

$$L = \frac{\phi_0}{2\pi I_C}\frac{1}{\cos(\varphi(t))} = L_J\left(\frac{1}{\cos(\varphi(t))}\right) \tag{8}$$

this shows that it is non-linear, which is what we want, so that the gaps between energy levels are not constant. Our next step is to find the expression for energy

$$\begin{aligned}
E &= \int_{t_A}^{t_B}(VI)\,dt \\
&= \int_{\varphi_A}^{\varphi_B}(V_{AB}\,I)\frac{dt}{d\varphi}\,d\varphi \\
&= \int_{\varphi_A}^{\varphi_B}(\cancel{V_{AB}}\,I)\frac{\phi_0}{2\pi\cancel{V_{AB}}}\,d\varphi \\
&= \frac{\phi_0}{2\pi}\int_{\varphi_A}^{\varphi_B} I_C\sin(\varphi)\,d\varphi \\
&= \left(\frac{I_C\phi_0}{2\pi}\right)[\cos(\varphi_A) - \cos(\varphi_B)] \\
E &= E_J[\cos(\varphi_A) - \cos(\varphi_B)]
\end{aligned} \tag{9}$$

Taking the integration limits to be 0 to φ, the energy expression becomes

$$E = E_J(1 - \cos\varphi) \tag{10}$$

$$E = E_J\left(\frac{\varphi^2}{2} - \frac{\varphi^4}{4!} + ...\right)$$

One can see how it is close to the harmonic oscillator but not exactly. There is some non-linearity, which is what we want. The Josephson energy E_J and Josephson inductance L_J are related by

$$E_J = \left(\frac{\phi_0}{2\pi}\right)^2 \left(\frac{1}{L_J}\right) \tag{11}$$

Now, for proper biasing when fabricating this setup, one must be able to get the specific critical current I_C required for the qubit's operating frequency. It will depend on Δ, but that is fixed for a material, so we cannot adjust it. Thus, the critical current is controlled by the *normal state resistance* R_n. It is the resistance of the junction, if the aluminium part was not superconducting (hence "normal"). The relation is given by

$$I_C = \frac{\pi\Delta}{2R_n e} \tag{12}$$

R_n can be adjusted by changing one of these two parameters:

- Oxidation time, i.e., the time period for which the aluminium is exposed to oxygen. Increasing this time means a thicker AlO_x region is formed, leading to a larger R_n.
- Junction area. Increasing this leads to a lower R_n, as expected.

1.3 Making a Qubit out of the Josephson Junction

1.3.1 Making an Oscillator

Now we have established that the Josephson junction works as a non-linear inductor. On its own, it doesn't mean much, it just charges up to a certain energy and nothing else. We need it to be in some sort of oscillation. Hence, we bring in a capacitor to make it an LC oscillator, as shown in Figure 4.

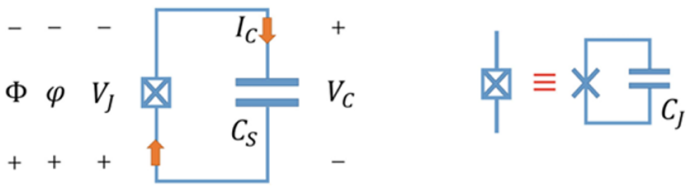


Figure 4: LC Oscillator made using Josephson junction and Shunt Capacitance C_S. The built-in capacitance C_J is small compared to C_S, so it can be ignored. [1]

As we can see, the capacitance C_S combines with the Josephson inductance to form a non-linear LC oscillator (an "artificial atom"). This creates discrete and measurable energy levels with a transition frequency. Without the capacitor, the system would not have a "plasma frequency" (natural frequency).
Now, to describe the system quantum mechanically, we write its Hamiltonian, as the sum of the energy stored in the capacitor and the inductor

$$H = \frac{Q^2}{2C_S} + E_J(1 - \cos\varphi) \tag{13}$$

We need to write this in operator form. The only things that change with time are the charge Q and the phase φ. So those will be the operators. As constants in a Hamiltonian do not contribute anything, we can ignore the constant E_J part as well.

$$\hat{H} = \frac{\hat{Q}^2}{2C_S} - E_J\cos\hat{\varphi} \tag{14}$$

We can rewrite the charge operator $\hat{Q}$ in terms of the number operator of Cooper pairs, as $\hat{Q} = 2e\hat{n}$. **Note that** $\hat{n}$ does not refer to the *total* number of Cooper pairs in the superconductors, just the ones that tunnel through the junction and travel through the circuit to C_S. The equation then becomes

$$\begin{aligned}\hat{H} &= \frac{(2e\hat{n})^2}{2C_S} - E_J\cos\hat{\varphi} \\ &= 4\left(\frac{e^2}{2C_S}\right)\hat{n}^2 - E_J\cos\hat{\varphi} \\ \hat{H} &= 4E_C\hat{n}^2 - \frac{1}{2}E_J\left(e^{i\hat{\varphi}} + e^{-i\hat{\varphi}}\right)\end{aligned} \tag{15}$$

Looking at the Hamiltonian expression, one can notice the parallels between the operators $\hat{n}$ and $\hat{\varphi}$ and the operators $\hat{p}$ and $\hat{x}$. We say $\hat{\varphi}$ and $\hat{n}$ are the conjugate variables of the system, and much like the position and momentum operators, they follow the commutation relation

$$[\hat{\varphi}, \hat{n}] = i \tag{16}$$

We now wish to find the commutation relation between $\hat{n}$ and $e^{i\hat{\varphi}}$. To find it, we use the general identity that if $[\hat{n}, \hat{\varphi}] = -i$, then for any analytical function $f(\hat{\varphi})$,

$$[\hat{n}, f(\hat{\varphi})] = -i\frac{\partial f}{\partial \hat{\varphi}} \tag{17}$$

This can be proved by verifying it for a power of the phase $\hat{\varphi}^m$, thus showing that it is true for any function that has a power series, i.e. analytical functions.
Now, using that relation,

$$\begin{aligned}
[\hat{n}, e^{i\hat{\varphi}}] &= -i\frac{\partial e^{i\hat{\varphi}}}{\partial \hat{\varphi}} \\
&= -i \cdot i\, e^{i\hat{\varphi}} \\
[\hat{n}, e^{i\hat{\varphi}}] &= e^{i\hat{\varphi}} \qquad (18) \\
\Rightarrow \hat{n}e^{i\hat{\varphi}} - e^{i\hat{\varphi}}\hat{n} &= e^{i\hat{\varphi}} \\
\Rightarrow \hat{n}e^{i\hat{\varphi}} &= e^{i\hat{\varphi}}(\hat{n} + \hat{\mathbb{1}}) \qquad (19)
\end{aligned}$$

Now we introduce the Cooper pair number eigenstates $|n\rangle$, by definition, they are eigenstates of their corresponding number operator with the eigenvalue equation

$$\hat{n}|n\rangle = n|n\rangle \tag{20}$$

Bringing them into the earlier equation, we get

$$\begin{aligned}
\hat{n}e^{i\hat{\varphi}}|n\rangle &= e^{i\hat{\varphi}}(\hat{n} + \hat{\mathbb{1}})|n\rangle \\
&= e^{i\hat{\varphi}}(n+1)|n\rangle \\
\Rightarrow \hat{n}\big(e^{i\hat{\varphi}}|n\rangle\big) &= (n+1)\big(e^{i\hat{\varphi}}|n\rangle\big) \qquad (21)
\end{aligned}$$

This means $e^{i\hat{\varphi}}|n\rangle$ is an eigenstate of $\hat{n}$ with eigenvalue $(n+1)$. The only eigenstate that fits that description is $|n+1\rangle$. Hence,

$$e^{i\hat{\varphi}}|n\rangle = |n+1\rangle \tag{22}$$

This means $e^{i\hat{\varphi}}$ is the raising operator of these eigenstates. This will be useful soon.
Our next step is to find the matrix elements of the Hamiltonian.

$$\begin{aligned}
\langle m|\hat{H}|n\rangle &= 4E_C\langle m|\hat{n}^2|n\rangle - \frac{1}{2}E_J\langle m|e^{i\hat{\varphi}}|n\rangle - \frac{1}{2}E_J\langle m|e^{-i\hat{\varphi}}|n\rangle \\
&= 4E_C\, n^2\langle m|n\rangle - \frac{1}{2}E_J\langle m|n+1\rangle - \frac{1}{2}E_J\langle m|n-1\rangle \\
&= 4E_C\, n^2\,\delta_{m,n} - \frac{1}{2}E_J\,\delta_{m,n+1} - \frac{1}{2}E_J\,\delta_{m,n-1} \qquad (23)
\end{aligned}$$

Now writing the Hamiltonian in bra-ket notation, we find it to be

$$\hat{H} = \sum_{n=0}^{N-1}\left[4E_C\, n^2|n\rangle\langle n| - \frac{1}{2}E_J|n+1\rangle\langle n| - \frac{1}{2}E_J|n-1\rangle\langle n|\right] \tag{24}$$

Taking the two-level system of 0 and 1, as that is all we care about for qubits, we get

$$\begin{aligned}
\hat{H} &= \left(4E_C\,(0^2)|0\rangle\langle 0| - \frac{1}{2}E_J|1\rangle\langle 0|\right) + \left(4E_C\,(1^2)|1\rangle\langle 1| - \frac{1}{2}E_J|0\rangle\langle 1|\right) \\
\Rightarrow \hat{H} &= -\frac{1}{2}E_J|0\rangle\langle 1| - \frac{1}{2}E_J|1\rangle\langle 0| + 4E_C|1\rangle\langle 1| \qquad (25)
\end{aligned}$$

This truncation is suspect, because it assumes either 1 Cooper pair tunnels through the junction, or no pairs tunnel. This *could* be justified by making the argument that, as $4E_C\, n^2$ grows quadratically, the gap between energy levels increases as n increases, so more pairs tunnelling is less likely and can reasonably be ignored.
However, even this justification is not entirely honest. In reality, when realising this setup, we choose a very large C_S

(in comparison to built-in capacitance C_J) such that $E_C \ll E_J$ (the qubit realised this way is called a *Transmon qubit*). This is done in order to decrease the energy dependence on the number of pairs tunnelled, thus decreasing the systems sensitivity to *charge noise* (stray charges from the environment).
In the end, this truncation was only to get a Hamiltonian confined to our basis states, so we can show it as a 2×2 matrix. In reality, we consider a few higher-up states and then calculate the energy level gaps.
Thus, in matrix form, the Hamiltonian is

$$\hat{H} = \begin{bmatrix} 0 & -\frac{1}{2}E_J \\ -\frac{1}{2}E_J & 4E_C \end{bmatrix} \tag{26}$$

1.3.2 Finding the Energy Gaps

So now we have the Hamiltonian of our system. Our next task is to find the frequency ω_{01} required to cause a transition between levels, and see if it is different enough from ω_{12}. For this, we will work with the Hamiltonian

$$\hat{H} = 4E_C\hat{n}^2 + E_J(1 - \cos\hat{\varphi}) = 4E_C\hat{n}^2 + \frac{1}{2}E_J\hat{\varphi}^2 - \frac{1}{24}E_J\hat{\varphi}^4 \tag{27}$$

Our first step will be to find the frequency of the harmonic part. We will ignore the $\hat{\varphi}^4$ term, and compare it to the harmonic oscillator Hamiltonian

$$\hat{H}_{\text{HO}} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2\hat{x}^2 \tag{28}$$

We see the equivalent terms

$$\frac{1}{2m} \equiv 4E_C \qquad \frac{1}{2}m\omega^2 \equiv \frac{1}{2}E_J \tag{29}$$

We multiply these to find the plasma frequency ω_p of the oscillator:

$$\begin{aligned} \frac{1}{2m}\frac{1}{2}m\omega_p^2 &= 4E_C\frac{1}{2}E_J \\ \Rightarrow \omega_p &= \frac{1}{\hbar}\sqrt{8E_CE_J} \end{aligned} \tag{30}$$

The $\hbar$ was brought in at the end for dimensional correctness.
The energy levels will look like

$$E_m^{(0)} = \hbar\,\omega_p\left(m + \frac{1}{2}\right) \tag{31}$$

where $m = 0, 1, 2, ...$ We're using m so as not to confuse it with n, which refers to the Cooper pairs.
Now the perturbation Hamiltonian is $\hat{H}' = -\frac{1}{24}E_J\hat{\varphi}^4$. First-order perturbation theory states that the correction term $E_m^{(1)}$ is given by

$$E_m^{(1)} = \langle m|\hat{H}'|m\rangle = -\frac{E_J}{24}\langle m|\hat{\varphi}^4|m\rangle \tag{32}$$

which is the effect of the perturbation on the unperturbed eigenstate. Now we want to rewrite the phase in terms of the system's ladder operators. For a harmonic oscillator, the ladder operators are defined as

$$\hat{a} = \sqrt{\frac{m\omega_p}{2\hbar}}\hat{x} + \frac{i}{\sqrt{2m\omega_p\hbar}}\hat{p} \tag{33}$$

$$\Rightarrow \hat{x} = \sqrt{\frac{\hbar}{2m\omega_p}}(\hat{a} + \hat{a}^\dagger) \tag{34}$$

Using $\hat{x} \equiv \hat{\varphi}$ and (29), we get

$$\hat{\varphi} = \left(\frac{2E_C}{E_J}\right)^{1/4}(\hat{a} + \hat{a}^\dagger) = \varphi_0(\hat{a} + \hat{a}^\dagger) \tag{35}$$

Substituting this into (32), we get

$$E_m^{(1)} = -\frac{E_J}{24}\left(\frac{2E_C}{E_J}\right)\langle m|(\hat{a} + \hat{a}^\dagger)^4|m\rangle = -\frac{E_C}{12}\langle m|(\hat{a} + \hat{a}^\dagger)^4|m\rangle \tag{36}$$

Now expanding $(\hat{a} + \hat{a}^\dagger)^4$ will give 16 terms, but we can simplify this by remembering the action of these ladder operators

$$\hat{a}|m\rangle = \sqrt{m}|m-1\rangle \qquad \hat{a}^\dagger|m\rangle = \sqrt{m+1}|m+1\rangle \tag{37}$$

Thus, we can say, for integers p and q,

$$\hat{a}^p \hat{a}^{\dagger q}|m\rangle = c|m-p+q\rangle \tag{38}$$

$$\begin{aligned}\langle m|\hat{a}^p \hat{a}^{\dagger q}|m\rangle &= c\langle m|m-p+q\rangle \\ &= 0, \text{ unless } p=q\end{aligned} \tag{39}$$

Therefore, the only terms of $(\hat{a}+\hat{a}^\dagger)^4$ that will survive are the ones with equal weight of $\hat{a}$ and $\hat{a}^\dagger$.
Those are: $\hat{a}^2\hat{a}^{\dagger 2} + \hat{a}^{\dagger 2}\hat{a}^2 + \hat{a}^\dagger\hat{a}^2\hat{a}^\dagger + \hat{a}\hat{a}^{\dagger 2}\hat{a} + \hat{a}\hat{a}^\dagger\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a}$.
Substituting this into the $E_m^{(1)}$ equation:

$$\begin{aligned}E_m^{(1)} &= -\frac{E_C}{12}\langle m|\hat{a}^2\hat{a}^{\dagger 2} + \hat{a}^{\dagger 2}\hat{a}^2 + \hat{a}^\dagger\hat{a}^2\hat{a}^\dagger + \hat{a}\hat{a}^{\dagger 2}\hat{a} + \hat{a}\hat{a}^\dagger\hat{a}\hat{a}^\dagger + \hat{a}^\dagger\hat{a}\hat{a}^\dagger\hat{a}|m\rangle \\ &= -\frac{E_C}{12}[(m+2)(m+1) + (m+1)^2 + m(m+1) + (m+1)m + m^2 + m(m-1)]\langle m|m\rangle \\ &= -\frac{E_C}{12}[m^2+3m+2+m^2+2m+1+m^2+m+m^2+m+m^2+m^2-m] \\ E_m^{(1)} &= -\frac{E_C}{12}[6m^2+6m+3]\end{aligned} \tag{40}$$

Therefore, the final expression for energy, upto first order, is

$$E_m = \hbar\omega_p\left(m+\frac{1}{2}\right) - \frac{E_C}{12}(6m^2+6m+3) \tag{41}$$

where $m = 0, 1, 2, ...$
Now to find the transition frequency from ground to first excited state:

$$\begin{aligned}\hbar\omega_{01} &= E_1 - E_0 \\ &= \hbar\omega_p(1-0) - \frac{E_C}{12}((6+6+3)-(0+0+3)) \\ \therefore \hbar\omega_{01} &= \sqrt{8E_C E_J} - E_C\end{aligned} \tag{42}$$

Now let's see what ω_{12} is:

$$\begin{aligned}\hbar\omega_{12} &= E_2 - E_1 \\ &= \hbar\omega_p(2-1) - \frac{E_C}{12}((24+12+3)-(6+6+3)) \\ \therefore \hbar\omega_{01} &= \sqrt{8E_C E_J} - 2E_C\end{aligned} \tag{43}$$

Therefore, the gaps between the energy levels differs by E_C. Since reducing E_C provides us with lower noise sensitivity, we have a trade-off. The middle ground chosen is usually $E_C \equiv 200 - 300\,\text{MHz}$. This corresponds to a shunt capacitance of around 100 fF, which may not seem like much, but when compared to $C_J \sim 1\,\text{fF}$, it is large enough.

1.4 Performing Qubit Operations

To perform qubit operations (logic gates), we provide a driving pulse at frequency ω_{01}. When this pulse is continuously applied, the qubit does not simply transition from $|0\rangle$ to $|1\rangle$. Instead, it oscillates between the two states *(Rabi oscillations)*. The probability of finding the qubit in $|1\rangle$ state is given by

$$P_{|1\rangle}(t) = \sin^2\left(\frac{\omega_R\, t}{2}\right) \tag{44}$$

where ω_R is the Rabi frequency, determined by the pulse intensity and coupling strength with the system. The pulse is of the form

$$\Omega(t) = \Omega_0 \sin(\omega_{01}t + \theta) \tag{45}$$

Adjusting the phase θ, we can adjust our axis of rotation, and by controlling the pulse duration, we can adjust the rotation angle. Thus, we can perform arbitrary rotations on the Bloch sphere, which is exactly what logic gates do.

1.5 Measuring the Qubit State

To measure the qubit's state, the regular method would be to send in a microwave pulse, and if it is of the energy level's resonant frequency, we will get a more dramatic response due to resonance. This cannot be done in our case, as there is a chance that the qubit just absorbs the photon and transitions to some other energy level, thus changing its state.
To get around this, we couple the system to a *readout resonator*, which is made with a meandered (twisting) coplanar waveguide. It has a well-defined resonant frequency ω_r, and couples to our system in a known way (say, through a known coupling capacitance). The resonant frequency of this system (qubit + resonator) will shift depending on the qubit state: it will be ω_r if the qubit is $|0\rangle$, and $\omega_r - \chi$ if the qubit is $|1\rangle$. Here, χ is called the dispersive shift. It does this by adding a term to the Hamiltonian, making it $\hbar(\omega_r - \chi\hat{a}^\dagger\hat{a})$, where $\hat{a}$ is the annihilation operator of the qubit.
Thus, if the qubit is in $|0\rangle$ state, the operation $\hat{a}^\dagger\hat{a}$ on it will give 0, thus the resonant frequency is ω_r. If the qubit is in $|1\rangle$ state, then the operation $\hat{a}^\dagger\hat{a}$ on it will give 1, thus the resonant frequency is $\omega_r - \chi$.
This system as a whole (qubit + resonator) is coupled to a probe. We can send in a signal at a frequency in between the two, say $\omega_r - \chi/2$, and measure the phase of the signal coming back. The direction of phase shift tells us whether the qubit is in $|0\rangle$ or $|1\rangle$ state.

- $|0\rangle$ state $\rightarrow$ ω_r resonant frequency $\rightarrow$ probing below resonance $\rightarrow$ positive phase shift
- $|1\rangle$ state $\rightarrow$ $\omega_r - \chi$ resonant frequency $\rightarrow$ probing above resonance $\rightarrow$ negative phase shift

Thus, we are able to measure the qubit state.

2 Trapped Ion Qubit

Trapped ion qubits are realised by isolating individual atomic ions within a vacuum chamber using electromagnetic fields. We need it to be ions because if they are neutral, they will simply drift through its surroundings. An ideal trapping potential for the ion is shown in Figure 5.

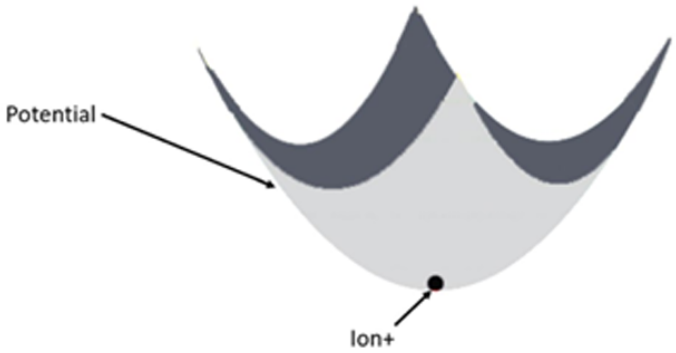


Figure 5: Ion trapped in Electrostatic Potential [1]

This trapping idea is made difficult thanks to *Earnshaw's Theorem*, which states that a collection of point charges cannot be maintained in a stable stationary equilibrium configuration solely by the interaction of electrostatic forces. We can see that this is due to Laplace's equation, where the potential must satisfy $\nabla^2\Phi = 0$. Thus the potential Φ is a harmonic function, meaning it satisfies the property that the average of Φ over a disk is equal to its value at the centre. This is clearly not true in the case of Figure 5. So there will always be at least one direction where the ion is unstable and can escape. We must figure out how to overcome this.
To get past the restrictions of Earnshaw's Theorem, we abandon purely static fields and work with a time-varying potential instead. This is called the *Paul Trap*, and it uses a combination of DC voltage and oscillating radio-frequency (RF) voltage like so:

$$\Phi(x, y, t) = \frac{V_{DC} + V_{RF}\cos(\omega_{RF}\, t)}{2r_0^2}(x^2 - y^2) \tag{46}$$

Here, V_{DC} is the static voltage, V_{RF} is the amplitude of the oscillating voltage, ω_{RF} is the frequency of oscillation and r_0 is the distance from the trap centre to the electrodes.
At any given moment, the potential will look like a saddle potential, where the ion can escape along a certain direction. However, the rapid oscillation means that before the ion can escape, the field reverses and pushes it back. This creates an effective trapping potential.

Now for an ion of charge e, the force on it is given by

$$F = m\ddot{r} = -e\nabla\Phi \tag{47}$$

Substituting the expression for potential into this equation and after working through the algebra, we obtain the equations of motion for x and y.

$$\frac{d^2x}{d\xi^2} + (a_x - 2q_x\cos(2\xi))x = 0 \tag{48}$$

$$\frac{d^2y}{d\xi^2} + (a_y - 2q_y\cos(2\xi))y = 0 \tag{49}$$

where $\xi = \frac{\omega_{RF}\,t}{2}$ is the dimensionless equivalent of time, used here. These equations are in the form of the **Mathieu Equation**, a linear DE where the coefficient also has time-dependence. The *stability parameters* a and q are defined as

$$a_x = \frac{4eV_{DC}}{m\omega_{RF}^2 r_0^2} \qquad q_x = -\frac{2eV_{RF}}{m\omega_{RF}^2 r_0^2} \tag{50}$$

$$a_y = -\frac{4eV_{DC}}{m\omega_{RF}^2 r_0^2} \qquad q_y = \frac{2eV_{RF}}{m\omega_{RF}^2 r_0^2} \tag{51}$$

The ion is successfully trapped only if these parameters fall within certain specific "stability regions" of the equation solution space. Within these regions, the ion has bounded motion. Thus, a combination of AC and DC fields can create a stable well for the ion to be trapped in.

A valid question at this point is to ask how the ions are created, and how one makes sure that only one ion gets trapped in the Paul Trap. A small oven heats up the element until it enters into a gaseous state and its atoms escape individually. Then a weak ionisation laser is applied to the sample. It is weak to make sure ionisation only happens occasionally. When one atom has been ionised, it will get trapped by the field.

We've created the system to realise qubits, our next step now is to define what a $|0\rangle$ and $|1\rangle$ state is. There are two ways: using electronic levels which have transition frequencies in the optical range, or hyperfine sublevels in the ground state that have microwave transition frequencies. In either case, the bottom two levels are labelled $|0\rangle$ and $|1\rangle$ respectively. Common ions used are Calcium for its simple electronic energy level structure, and Ytterbium, for its simple hyperfine structure.

To perform operations on the qubits, we use laser/microwave pulses, tuned to the transition (resonance) frequency. The same concept from superconducting qubits is used here, where the pulse duration allows for arbitrary rotations on the Bloch sphere, thus enabling the execution of single-qubit gates.

To measure qubits, we use a technique called *electron shelving*. A laser is tuned so that it causes a cycling transition at $|0\rangle$ — a transition from $|0\rangle$ to a short-lived auxiliary state. These auxiliary states can be found in the energy level structure of the ion. There are two possible cases:

1. *The ion is in* $|0\rangle$ *state:* In this case, the laser will cause the ion to transition to the auxiliary state, which then quickly decays back to the $|0\rangle$ state, emitting a photon. We call it a "cycling" transition because we want this process to repeat many times a second, thus producing noticeable light.
2. *The ion is in* $|1\rangle$ *state:* In this case, the laser will not cause any transition and no light will be detected.

Thus, we are able to easily tell which state the qubit is in. If we are using the hyperfine states, we should be more careful tuning the laser, as we could easily excite both states.

3 Photonic Qubit

Photonic qubits store quantum information using photons. In this approach, the quantum states $|0\rangle$ and $|1\rangle$ are represented by specific properties of the photon, such as polarisation, optical path (path encoding), time-bin or frequency.

- *Polarisation:* Take 2 orthogonal directions. One of them can be labelled as $|0\rangle$ and the other as $|1\rangle$. A polarisation direction in between these two will be a superposition state.
- *Optical Path:* Here, $|0\rangle$ refers to the photon being in the upper path, while $|1\rangle$ refers to the photon being in the lower path. This method is used in Photonic IC's (PIC's). Beam splitters can be used to make superposition states.
- *Time-bin:* Here the state is characterised by *when* the photon arrives. You divide time into two narrow windows: an "early" bin and a "late" bin. If the photon arrives early, it is in $|0\rangle$ state, and if it arrives late, it is in $|1\rangle$ state. We can use a Mach-Zehnder Interferometer (MZI) with different path lengths to obtain superposition states.

- *Frequency:* We can choose two distinct frequencies to be our $|0\rangle$ and $|1\rangle$ states.

We will use path encoding, as it allows us to realise these qubits on PIC's.
We can perform logical operations on the qubits using beam splitters and MZI's. The components of a MZI are shown in Figure 6

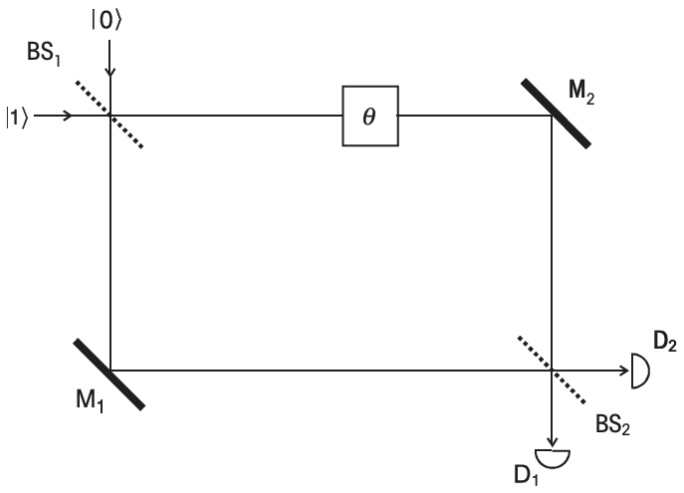


Figure 6: Mach-Zehnder Interferometer (MZI). The beam splitter BS_1 and BS_2 are 50:50. M_1 and M_2 are mirrors. D_1 and D_2 are detectors. θ is a phase shifter.[2]

By adjusting the phase shift θ, we can realise different logical gates. It is not as smooth as previous realisations, where simply by controlling pulse duration, one could perform any logical operation. Here, we need to solve backwards from the gate we want to realise, to calculate the θ required. Some examples are:

- Z gate can be realised by putting $\theta = \pi$.
- X gate can be realised by putting $\theta = 0$.

Both of these realisations are upto a global phase.
We can look into how this is derived. We know from our earlier learnings that the unitary matrix corresponding to a beam splitter is

$$U_{BS} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} \tag{52}$$

And the phase shifter matrix, denoted by $P(\theta)$ is

$$P(\theta) = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix} \tag{53}$$

Therefore, the general unitary realised by an MZI is given by

$$\begin{aligned} U &= U_{BS}\, P(\theta)\, U_{BS} \\ &= \frac{1}{2} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{bmatrix} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} \\ &= \frac{1}{2} \begin{bmatrix} 1 & i \\ i & 1 \end{bmatrix} \begin{bmatrix} 1 & i \\ ie^{i\theta} & e^{i\theta} \end{bmatrix} \\ \therefore U &= \frac{1}{2} \begin{bmatrix} 1 - e^{i\theta} & i(1 + e^{i\theta}) \\ i(1 + e^{i\theta}) & -1 + e^{i\theta} \end{bmatrix} \end{aligned} \tag{54}$$

There is actually an overall minus sign contributed by the mirrors, but that is ignored as it is an overall phase.
We can see now that choosing $\theta = \pi$ gives us the Z gate and $\theta = 0$ gives us iX.
Measurement of the qubit state is simply done by placing a detector in each path. If the upper one clicks, it is in $|0\rangle$, and if the lower one clicks, it is in $|1\rangle$.
A major advantage of photonic qubits is their low interaction rate with their environment, leading to much longer coherence times compared to the other two methods. Furthermore, they can be realised at room temperature, unlike superconducting

qubits, which makes them ideal for larger-scale projects.
However, challenges remain in creating photon sources that release single photons on demand. Realising two-qubit gates is another obstacle that is difficult to overcome, as photons don't really interact with each other.

4 Spin-based Qubit

In spin-based qubits, quantum information is encoded in the intrinsic spin angular momentum of a single electron. The logical states $|0\rangle$ and $|1\rangle$ correspond to the orthogonal spin states of the fermions: spin-down $|\downarrow\rangle$ and spin-up $|\uparrow\rangle$ respectively. To understand spin, let us first take the classical case. We consider a charge q moving in a circular path of radius R with velocity v, as shown in Figure 7.

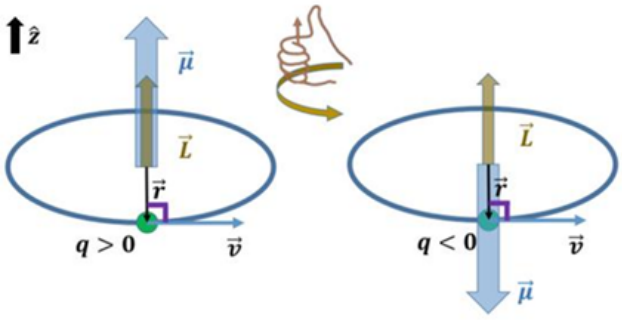


Figure 7: Visualisation of Angular Momentum [1]

The effective current I generated by this motion is

$$I = \frac{q}{T_{\text{cycle}}} = \frac{q}{\left(\frac{2\pi R}{v}\right)} = \frac{\frac{q}{m}(mv)}{2\pi R} = \frac{q}{2m}\frac{p}{\pi R} \tag{55}$$

The magnetic moment $\vec{\mu}$ is the product of this current and the area of the loop, in the direction given by the Right-Hand Rule.

$$\vec{\mu} = I \cdot A \cdot \hat{n} = \left(\frac{q}{2m}\frac{p}{\pi R}\right)(\pi R^2)\hat{z} = \frac{q}{2m}(pR)\hat{z} = \frac{q}{2m}\vec{L} \tag{56}$$

where $\vec{L}$ is the classical angular momentum.
In the quantum regime, L is replaced by the spin operator S, so we can define

$$\vec{S} = S\hbar\hat{z} \tag{57}$$

The magnetic moment $\vec{\mu}_e$ of an electron is given by

$$\vec{\mu}_e = g\frac{e}{2m}\vec{S} = g\frac{e\hbar}{2m}S\hat{z} = g\mu_B S\hat{z} \tag{58}$$

where g is the Lande g-factor, which is ≈ 2 for an electron, and μ_B is the Bohr magneton.
Now in a regular setting, two electrons with opposite spins can be in the same energy level (spin degeneracy). But when an external magnetic field $\vec{B}$ is applied, this degeneracy is lifted (Zeeman effect) as they have different magnetic moments. This splitting is shown in Figure 8.

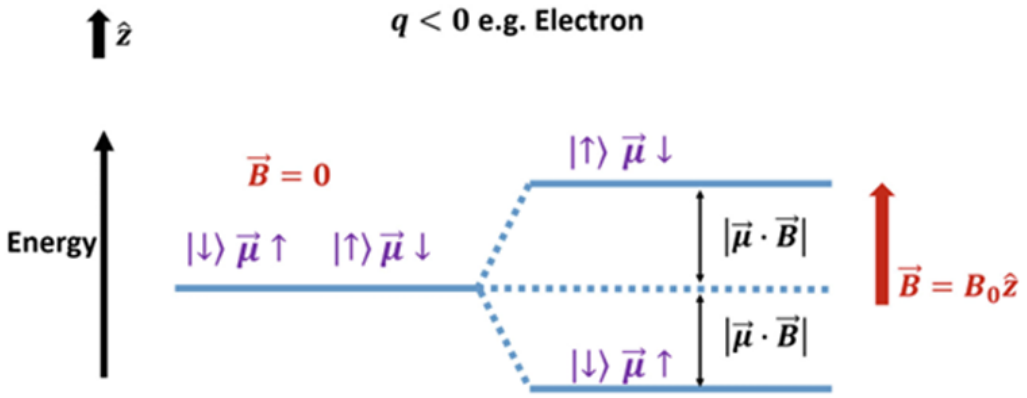


Figure 8: Zeeman Effect for Electron [1]

The two energy levels created are

$$E_{\text{spin}} = \begin{cases} -\frac{B_0 g \mu_B}{2} & \text{for spin-up } |\uparrow\rangle \\ \frac{B_0 g \mu_B}{2} & \text{for spin-down } |\downarrow\rangle \end{cases} \tag{59}$$

Thus, the Hamiltonian is

$$\hat{H} = \frac{B_0 g \mu_B}{2}|\downarrow\rangle\langle\downarrow| - \frac{B_0 g \mu_B}{2}|\uparrow\rangle\langle\uparrow| = \frac{B_0 g \mu_B}{2}|0\rangle\langle 0| - \frac{B_0 g \mu_B}{2}|1\rangle\langle 1| \tag{60}$$

In matrix form, we can see that it is proportional to the Pauli Z operator.

$$\hat{H} = \begin{bmatrix} \frac{B_0 g \mu_B}{2} & 0 \\ 0 & -\frac{B_0 g \mu_B}{2} \end{bmatrix} = \frac{B_0 g \mu_B}{2}\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \frac{B_0 g \mu_B}{2}\hat{Z} \tag{61}$$

Thus, we have a two-level system where we can easily label our basis states.
We can perform single qubit operations in the same way as the superconducting and trapped-ion qubits, where a microwave pulse is applied, and its duration is controlled to perform any operation.

4.1 Realisation using Quantum Dots

One way to physically realise these qubits is by using semiconductor structures called *quantum dots*. They act as artificial atoms by using electrostatic gates to create a potential well, confining an electron in three dimensions. This confinement allows the electron's spin to be controlled individually using microwave pulses.
Two-qubit operations can also be done using the setup illustrated in Figure 9.

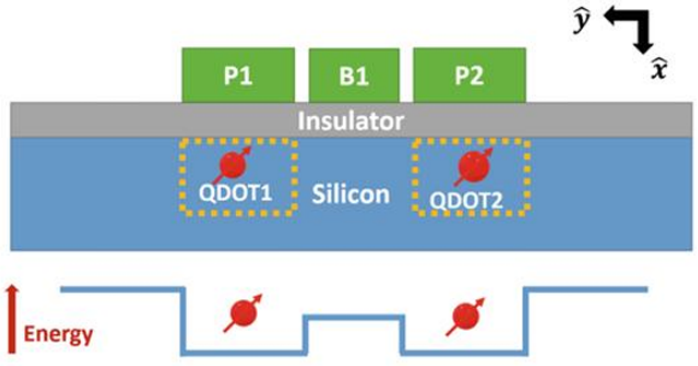


Figure 9: Quantum Dot Realisation for two qubits

Here, the "Plunger gates" P1 and P2 apply a potential that lowers the energy in specific regions, causing the electrons to be trapped in "potential wells". The "Barrier gate" B1 controls the height of the potential barrier between electrons. Lowering or raising this barrier affects the tunnelling probability, thus controlling how these qubits interact. This allows

for two-qubit gates to be realised.
Measurement of these qubits is done using a technique called *spin-to-charge conversion*. It is difficult to detect the magnetic moment of an electron directly. To get around this, we take advantage of the fact that the different states are at different energies, and we use the Plunger gate to provide an energy in between the two levels. If the qubit is in $|1\rangle$ state (higher energy), it will tunnel out, and if it is in $|0\rangle$ state (lower energy), it will not. It is called spin-to-charge conversion as it shows the spin information as a spatial movement of charge (electron).

4.2 Realisation using Nitrogen Vacancy (NV) Centres

Another method of realisation of spin-based qubits is using NV centres. These are point defects in the diamond lattice structure. It is a two-part defect consisting of:

1. A nitrogen atom replacing a carbon atom in the lattice
2. A vacant lattice site adjacent to the nitrogen atom

This can be seen in Figure 10.

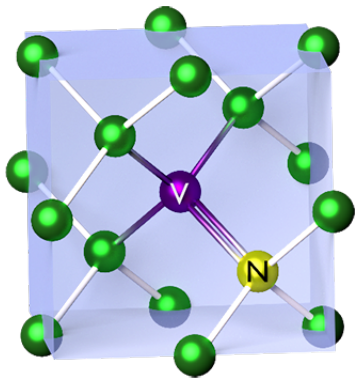


Figure 10: NV centre in a Diamond Lattice. Green spheres are Carbon atoms. 'N' is Nitrogen atom, and 'V' is the vacant lattice site [3]

4.2.1 Making of NV Centres

These NV centres are created by first firing nitrogen ions at the diamond to replace carbon atoms in the lattice (can also synthesise the material with a high concentration of nitrogen impurities), and then irradiating it with high-energy particles such as electrons, protons or other ions to displace the neighbouring carbon atom to create a vacancy. This forms a neutral NV^0 centre.
Electrons will be trapped in this vacancy. The trapping happens because the nitrogen vacancy distorts the lattice, forming a new set of electronic states that are confined to the vacancy. These levels are lower than the energy levels of the remaining lattice, which means the electron will not escape unless sufficiently excited. Thus, it is trapped.
Referring in Figure 10, we can see that the vacancy altogether contains 5 electrons that are not involved in a bond: 2 from nitrogen's lone pair, and 3 from the dangling carbon bonds. Two of those dangling electrons pair up to form a quasi-covalent bond between carbons, leaving 1 unpaired electron.
Now theoretically, our job is done here. We've isolated a single electron that is trapped in the vacancy, and we can manipulate it with microwave pulses like we did in the quantum dot case. We can also, in theory, measure the state using spin-to-charge conversion like before. However, that process was done by using gate voltages to create an energy level in between the two spin levels. Attaching wires and metal gates is unfeasible in a diamond; we need all our process to be optics-based.
To make this possible, we'll need to make NV^- centres. We make these by using donor ions to provide electrons to the centres. Since the vacancy already acts as a trap for electrons, it is not hard to get another electron trapped in it. Now there is a pair of electrons in the vacancy. This creates a different energy level structure, shown in Figure 11. We will now discuss it

4.2.2 Energy Level Structure

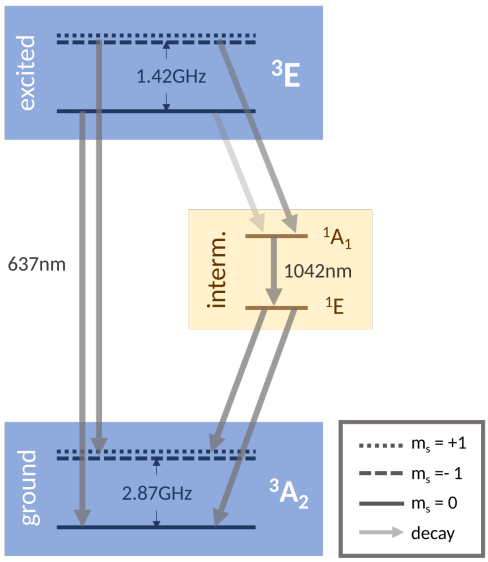


Figure 11: Energy level structure of NV$^-$ centre. The superscripts refer to the spin multiplicity $(2S+1)$. There are singlet $(S=0)$ and triplet $(S=1)$ states. The subscripts have to do with the symmetry of the wavefunctions, and are not relevant to us. The values shown are for type 1b diamond at room temperature. They are experimentally obtained.

Refer to Figure 11.
The NV centre has a ground state triplet $^3A_2$, an excited state triplet ^3E, and two intermediate state singlets $^1A_1$ and ^1E. The triplet states, naturally, have three states within them: $m_s = 0, +1, -1$. You may be wondering why both the $+1$ and -1 states are above the 0 state, as that goes against what we've learnt from the Zeeman effect. This is because the splitting seen here is *not due to the Zeeman effect*. There is no external magnetic field applied. The splitting is instead due to a mechanism known as *Zero-Field Splitting (ZFS)* [4].
ZFS occurs due to the spin-spin interaction between the two electrons in the NV centre. This interaction can be written in the form of a tensor $\bar{D}$. We assume the $\bar{D}$ tensor to be diagonal and traceless – diagonal because we are operating in its eigenbasis, and traceless because it is purely anisotropic (it only depends on relative spin orientations, not absolute). The interaction Hamiltonian is given by

$$\hat{H}^{\text{s-s}} = \hbar^{-2}\left[D\left(\hat{S}_z^2 - \frac{1}{3}\hat{S}^2\right) + E\left(\hat{S}_x^2 + \hat{S}_y^2\right)\right] \tag{62}$$

D here is "axial ZFS parameter" and E is the "rhombic ZFS parameter". They are mathematically defined as follows:

$$D = \frac{1}{2}(-D_{xx} - D_{yy} + 2D_{zz}) = \frac{3}{2}D_{zz} \qquad E = \frac{1}{2}(D_{xx}^2 - D_{yy}^2) \tag{63}$$

We know the definition of the spin operators in a spin-1/2 system $\left(\hat{S}_i = \frac{\hbar}{2}\hat{\sigma}_i\right)$. But here we have 3 states $|-1\rangle, |0\rangle$ and $|+1\rangle$. So the spin operators are defined as the following 3×3 matrices:

$$\hat{S}_x = \frac{\hbar}{\sqrt{2}}\begin{bmatrix}0 & 1 & 0\\ 1 & 0 & 1\\ 0 & 1 & 0\end{bmatrix} \qquad \hat{S}_y = \frac{\hbar}{\sqrt{2}}\begin{bmatrix}0 & -i & 0\\ i & 0 & -i\\ 0 & i & 0\end{bmatrix} \qquad \hat{S}_z = \hbar\begin{bmatrix}1 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & -1\end{bmatrix} \tag{64}$$

$$\hat{S}^2 = \hat{S}_x^2 + \hat{S}_y^2 + \hat{S}_z^2 = 2\hbar^2\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix} = 2\hbar^2 I \tag{65}$$

These matrix forms are obtained by first deriving the matrix form of the spin raising and lowering operators $\hat{S}_+$ and $\hat{S}_-$ in the 3×3 case. We derive them by knowledge of their action

$$\hat{S}_\pm|s, m_s\rangle = \hbar\sqrt{s(s+1) - m_s(m_s \pm 1)}|s, m_s \pm 1\rangle \tag{66}$$

In our case, the NV centre has C_{3v} symmetry, which consists of invariance in the following operations (Refer Figure 10):

- 120° rotation about the N-V axis.
- Reflection in mirror planes containing the N-V axis and passing through any one of the carbon atoms.

Since this is an axial symmetry, the D_{xx} and D_{yy} values will be the same and thus $E = 0$. The Hamiltonian simplifies to

$$\hat{H}^{\text{s-s}} = \hbar^{-2} D\left(\hat{S}_z^2 - \frac{1}{3}\hat{S}^2\right) \tag{67}$$

Now if we define our basis states as column vectors:

$$|+1\rangle = \begin{bmatrix}1\\0\\0\end{bmatrix} \qquad |0\rangle = \begin{bmatrix}0\\1\\0\end{bmatrix} \qquad |-1\rangle = \begin{bmatrix}0\\0\\1\end{bmatrix} \tag{68}$$

We can see that the action of $\hat{S}_z^2$ and $\hat{S}^2$ are as follows:

$$\hat{S}_z^2|m_s\rangle = m_s^2\hbar^2|m_s\rangle \qquad \hat{S}^2|m_s\rangle = S(S+1)\hbar^2|m_s\rangle = 2\hbar^2|m_s\rangle \tag{69}$$

From this, we can see that the action is same whether $m_s = -1$ or $+1$. Let us find the energy for each case:

$$\epsilon_0 = \hbar^{-2} D\left(0^2\hbar^2 - \frac{1}{3}2\hbar^2\right)$$

$$\Rightarrow \epsilon_0 = -\frac{2D}{3} \tag{70}$$

$$\epsilon_{\pm 1} = \hbar^{-2} D\left((\pm 1)^2\hbar^2 - \frac{1}{3}2\hbar^2\right)$$

$$\Rightarrow \epsilon_{\pm 1} = \frac{D}{3} \tag{71}$$

$$\therefore \Delta\epsilon = D \tag{72}$$

Therefore, we can see that the $m_s = \pm 1$ levels are of higher energy than the $m_s = 0$ level by a separation of D.
This then causes a problem where, with the same transition frequency, we can excite an electron to either the $|-1\rangle$ or the $|+1\rangle$ state. We want to be able to control which state it goes to. To allow for this, we apply a small biasing magnetic field, which leads to a Zeeman effect splitting between the $|-1\rangle$ and $|+1\rangle$ states. It should be just large enough to allow for us to cause one transition without causing the other.
It is worth clarifying what exactly we mean by "excited" here. Since this energy level structure is caused by two electrons, one may think that they must be together at all times. Not in the same orbital of course, but at least in the same energy electronic energy level. Actually, that is not the case. The excited state refers to only one electron being excited.

4.2.3 One electron vs. Two electrons

This requirement of a magnetic field should cause us to pause and reconsider why we brought in a second electron at all. Earlier, it had the advantage of not needing an external magnetic field, but that is lost now. Is it just adding unnecessary complications to an otherwise simple setup?
Not really. Having a spin $S = 1$ system has its advantages over a regular spin $S = 1/2$ system.

1. **Optical Initialisation:** In a spin-1/2 system with one electron, the only way to improve the chance of all your qubits going to $|0\rangle$ state is by reducing temperature such that $k_B T \ll \hbar\omega_{01}$, which requires temperatures in millikelvins. At room temperature, there is a roughly equal population of spin-up and spin-down states.
   In our spin-1 system with two electrons, we are able to initialise our qubits to $|0\rangle$ with high probability using optical methods. We will discuss this in detail soon.
2. **Optical Readout/Measurement:** As mentioned before, to use the same spin-charge conversion method of measurement that was used in quantum dots, we would have to use wires and metal pads to provide the "in-between" energy level, which is tough for diamond. The spin-1 system provides us with an optical way of reading the qubit state, which we will discuss.

To properly understand how these two processes occur, we need to take a brief detour to understand symmetries and how they matter in electronic transitions.

4.2.4 Symmetries of the Energy Levels [5]

To reiterate, the NV centre has C_{3v} symmetry, which consists of invariance in the following operations (Refer Figure 10):

- 120° rotation about the N-V axis.
- Reflection in mirror planes containing the N-V axis and passing through any one of the carbon atoms.

In this kind of point group, there are three types of states, based on the symmetry shown by their wavefunctions:

- A_1 *state:* When rotated 120° about the N-V axis or when reflected in a mirror containing the principal axis, its wavefunction stays the same.
- A_2 *state:* When rotated 120° about the N-V axis, the sign of its wavefunction stays the same, but when reflected in a mirror containing the principal axis, its sign flips.
- E *state:* When rotated 120° about the N-V axis or when reflected in a mirror containing the principal axis, its new wavefunction does not have any proportional relation to the old wavefunction. It changes to a new expression. This group contains two states: E_x and E_y. Upon rotating one of them, say E_x, it becomes a mix of the two.

Spin-orbit (SO) coupling interestingly couples the spin $m_s = -1, 0, +1$ to the symmetry A_1, A_2, E_x, E_y, and this is the reason why the symmetry-dependent transitions end up having an implicit dependence on the spin as well. Because of the coupling, m_s is no longer a good quantum number, as we have different states with the same m_s. Refer Table 1 to see the relation between symmetry and spin, and note that it is not a simple one-to-one mapping.
Without SO coupling, crossing between singlet and triplet states, known as *Inter-System Crossing* (ISC) would not be possible.

4.2.5 State Initialisation

The $|0\rangle$ and $|1\rangle$ states of our qubit are typically chosen to be the $m_s = 0$ and $m_s = -1$ states respectively of the ground $^3A_2$ state. The transition frequency at zero bias field is 2.87 GHz. It will be adjusted based on the bias field used.
Now, we want a reliable way to be sure that our qubits are in the $|0\rangle$ state before we begin our operations. Most setups do this by lowering the temperature, but that has its own set of problems. In an NV^- centre, we can do this optically.
We excite the state with a laser of wavelength ≤ 637nm. This takes the electron to the ^3E state. From there, it has two possible decay paths (Refer Figure 11):

1. *Radiative:* Direct decay from ^3E to $^3A_2$, releasing a red photon (637 nm).
2. *Non-radiative:* Decay via the intermediate singlet states, releasing infrared photons (thus not visible).

The fact that they are radiative/non-radiative does not matter here, but will matter in the measurement process. Now let us refer to Table 1. There are two possible cases:

1. $|\mathbf{0}\rangle$: Starting from $m_s = 0$ state (A_1) in $^3A_2$, an electron is excited to $m_s = 0$ state (E) in ^3E.
   - It can decay radiatively back to its initial state, with its spin staying as $m_s = 0$.
   - It *cannot* decay non-radiatively via the intermediate states, as $^1A_1$ state has A_1 symmetry, and in ISC, $E \to A_1$ is not allowed.
   - Direct $^3E \to {}^1E$, though permitted by symmetry, does not happen due to the energy gap being too large for ISC.

   Thus, when $m_s = 0$ state electrons are excited, when they decay, they are still $m_s = 0$ state, i.e., $|0\rangle$ state.
2. $|\mathbf{1}\rangle$: Starting from $m_s = \pm 1$ state (E) in $^3A_2$, an electron is excited to $m_s = \pm 1$ state ($E/A_1/A_2$) in ^3E.
   - No matter the state symmetry, it can decay radiatively back to its initial state, staying in $m_s = \pm 1$ state.
   - The ones in A_1 or A_2 state can decay via the intermediate states by going to $^1A_1$, as it has A_1 symmetry. It then goes to the $^3A_2$ while conserving symmetry. So it stays in A_1, meaning it goes to $m_s = 0$ state.
   - Another possible path for them is $^3E \to {}^1A_1 \to {}^1E \to {}^3A_2$, which when conserving symmetry, takes it to the E states which have $m_s = \pm 1$.

The radiative path is still the most popular one for both cases. But since there exists a path that takes ± 1 states to 0 states, but no such paths that take 0 states to ± 1 states, we can keep repeating this excitation-decay loop until we are sure that $\sim 99\%$ of our qubits are in the $|0\rangle$ state.
Thus, we are able to initialise our qubits using optical methods, while staying in room temperature.

Symmetry Representation		m_s
w/o SO	SO	
^3E	E_x	±1
	E_y	±1
	E_x	0
	E_y	0
	A_2	±1
	A_1	±1
$^1A_1$	A_1	0
^1E	E_x	0
	E_y	0
$^3A_2$	E_x	±1
	E_y	±1
	A_1	0

Table 1: The states of the NV^- centre. Spin-Orbit (SO) coupling causes a distinction is state symmetries based on the spin. The actual wavefunction expressions were excluded for simplicity, but they can be seen in Table II of [5]. These are actually many-body states of the 6 electrons in 8 available orbitals, so the reality is more complex than we have described. However, this table contains all the information we will need.

4.2.6 Measurement Process

For the measurement process, we excite the state with a green laser (∼532 nm). This takes the electron to the ^3E state. From there, it has two possible decay paths (Refer Figure 11):

1. *Radiative:* Direct decay from ^3E to $^3A_2$, releasing a red photon (637 nm).
2. *Non-radiative:* Decay via the intermediate singlet states, releasing infrared photons (thus not visible).

As we discussed in the previous section, electrons excited from the $|0\rangle$ state can only decay radiatively, while those excited from the $|1\rangle$ state can either decay radiatively or non-radiatively.
Thus, the qubits in $|0\rangle$ state will emit brighter red light when excited with green light compared to qubits in the $|1\rangle$ state, which is how we can distinguish between the two.
We use a green laser, despite it having more energy than the transition requires, so that we can easily tell apart the red light emitted by the decay transition.

NV centres are very useful in quantum sensing, as super-sensitive magnetic field sensors [6]. Thanks to zero-field splitting combined with the Zeeman effect, the transition frequency for $|0\rangle \to |-1\rangle$ is different to the frequency for $|0\rangle \to |+1\rangle$. Subtracting the two frequencies will help you obtain the strength of the magnetic field.

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